

Database Systems: Design, Implementation, and Management

Eighth Edition

Chapter 13

Business Intelligence and Data Warehouses

Objectives

- In this chapter, you will learn:
 - How business intelligence is a comprehensive framework to support business decision making
 - How operational data and decision support data differ
 - What a data warehouse is, how to prepare data for one, and how to implement one
 - What star schemas are and how they are constructed

Objectives (continued):

- In this chapter, you will learn: (continued)
 - What data mining is and what role it plays in decision support
 - About online analytical processing (OLAP)
 - How SQL extensions are used to support OLAP-type data manipulations

The Need for Data Analysis

- Managers track daily transactions to evaluate how the business is performing
- Strategies should be developed to meet organizational goals using operational databases
- Data analysis provides information about short-term tactical evaluations and strategies

Business Intelligence

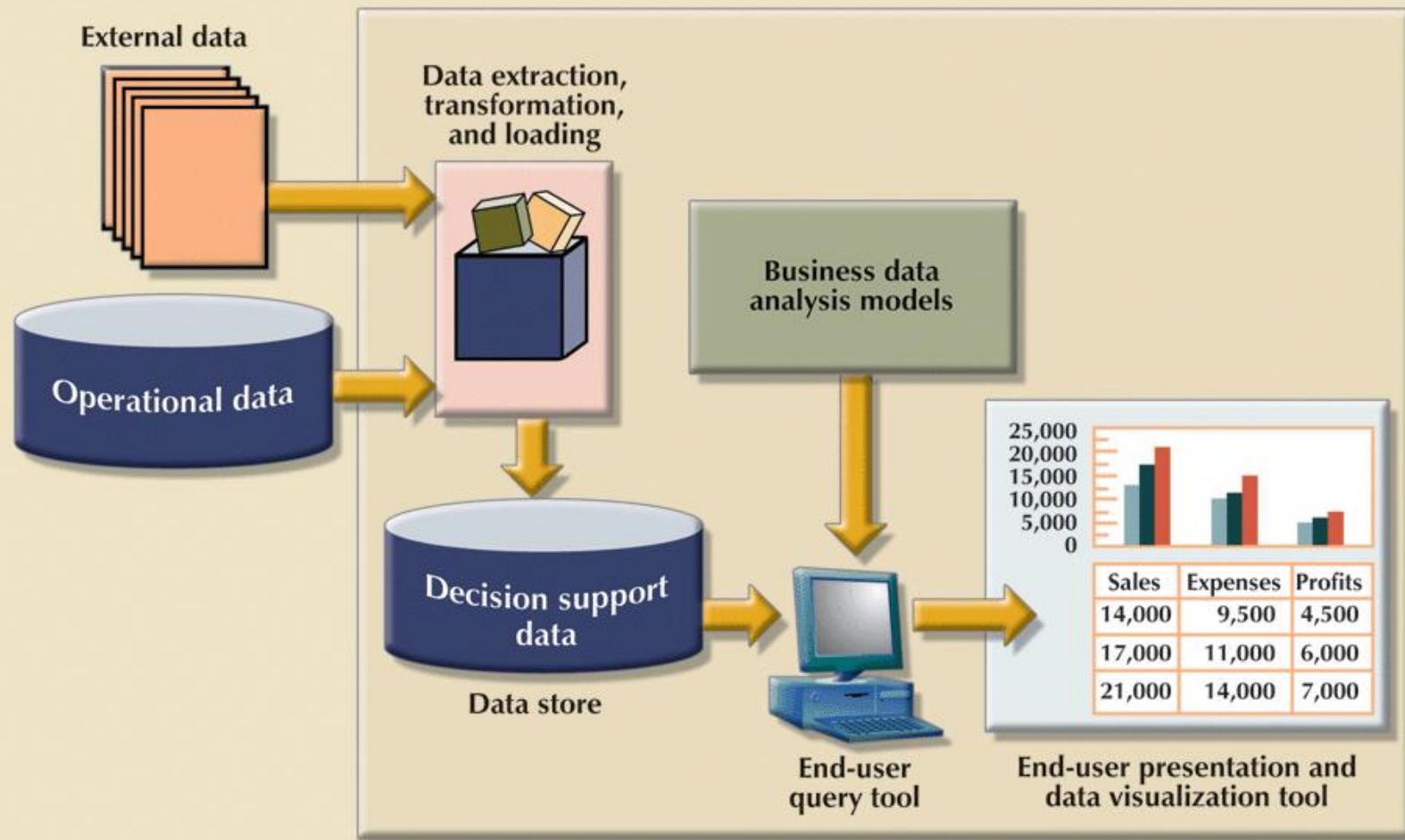
- Comprehensive, cohesive, integrated tools and processes
 - Capture, collect, integrate, store, and analyze data
 - Generate information to support business decision making
- Framework that allows a business to transform:
 - Data into information
 - Information into knowledge
 - Knowledge into wisdom

Business Intelligence Architecture

- Composed of data, people, processes, technology, and management of components
- Focuses on strategic and tactical use of information
- **Key performance indicators (KPI)**
 - Measurements that assess company's effectiveness or success in reaching goals
- Multiple tools from different vendors can be integrated into a single BI framework

**FIGURE
13.2**

Business intelligence components

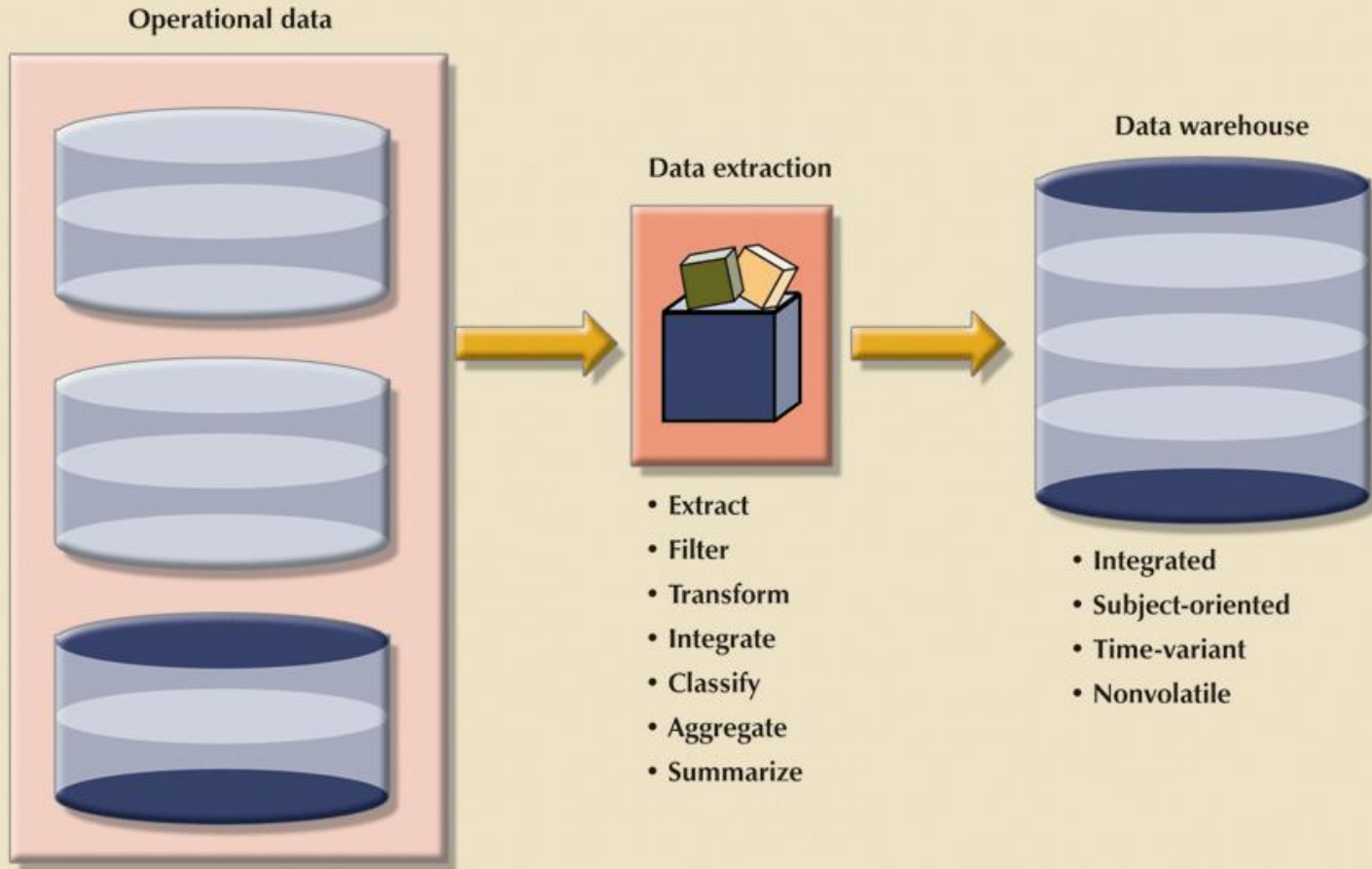


Decision Support Data

- Operational data
 - Mostly stored in relational database
 - Optimized to support transactions representing daily operations
- Decision support data differs from operational data in three main areas:
 - Time span
 - Granularity
 - Dimensionality

**FIGURE
13.4**

Creating a data warehouse



Decision Support Database Requirements

- Specialized DBMS tailored to provide fast answers to complex queries
- Four main requirements:
 - Database schema
 - Data extraction and loading
 - End-user analytical interface
 - Database size

Decision Support Database Requirements (continued)

- Database schema
 - Complex data representations
 - Aggregated and summarized data
 - Queries extract multidimensional time slices
- Data extraction and filtering
 - Supports different data sources
 - Flat files
 - Hierarchical, network, and relational databases
 - Multiple vendors
 - Checking for inconsistent data

Decision Support Database Requirements (continued)

- End-user analytical interface
 - One of most critical DSS DBMS components
 - Permits user to navigate through data to simplify and accelerate decision-making process
- Database size
 - In 2005, Wal-Mart had 260 terabytes of data in its data warehouses
 - DBMS must support **very large databases (VLDBs)**

The Data Warehouse

- Integrated, subject-oriented, time-variant, and nonvolatile collection of data
 - Provides support for decision making
- Usually a read-only database optimized for data analysis and query processing
- Requires time, money, and considerable managerial effort to create

The Data Warehouse (continued)

- **Data mart**
 - Small, single-subject data warehouse subset
 - More manageable data set than data warehouse
 - Provides decision support to small group of people
 - Typically lower cost and lower implementation time than data warehouse

Twelve Rules that Define a Data Warehouse

- Data warehouse and operational environments are separated
- Data warehouse data are integrated
- Data warehouse contains historical data over long time
- Data warehouse data are snapshot data captured at given point in time
- Data warehouse data are subject-oriented

Twelve Rules that Define a Data Warehouse (continued)

- Data warehouse data are mainly read-only
 - Periodic batch updates from operational data
 - No online updates allowed
- Data warehouse development life cycle differs from classical systems development
- Data warehouse contains data with several levels of detail:
 - Current detail data, old detail data, lightly summarized data, and highly summarized data

Twelve Rules that Define a Data Warehouse (continued)

- Read-only transactions to very large data sets
- Data warehouse environment traces data sources, transformations, and storage
- Data warehouse's metadata are critical component of this environment
- Data warehouse contains chargeback mechanism for resource usage
 - Enforces optimal use of data by end users

Decision Support Architectural Styles

- Provide advanced decision support features
- Some capable of providing access to multidimensional data analysis
- Complete data warehouse architecture supports:
 - Decision support data store
 - Data extraction and integration filter
 - Specialized presentation interface

Online Analytical Processing

- Advanced data analysis environment that supports:
 - Decision making
 - Business modeling
 - Operations research
- Four main characteristics:
 - Use multidimensional data analysis techniques
 - Provide advanced database support
 - Provide easy-to-use end-user interfaces
 - Support client/server architecture

Multidimensional Data Analysis Techniques

- Data are processed and viewed as part of a multidimensional structure
- Augmented by the following functions:
 - Advanced data presentation functions
 - Advanced data aggregation, consolidation, and classification functions
 - Advanced computational functions
 - Advanced data modeling functions

**FIGURE
13.5**

Operational vs. multidimensional view of sales

Database name: Ch13_Text

Table name: DW_INVOICE

INV_NUM	INV_DATE	CUS_NAME	INV_TOTAL
2034	15-May-08	Dartonik	1400.00
2035	15-May-08	Summer Lake	1200.00
2036	16-May-08	Dartonik	1350.00
2037	16-May-08	Summer lake	3100.00
2038	16-May-08	Trydon	400.00

Table name: DW_LINE

INV_NUM	LINE_NUM	PROD_DESCRIPTION	LINE_PRICE	LINE_QUANTITY	LINE_AMOUNT
2034	1	Optical Mouse	45.00	20	900.00
2034	2	Wireless RF remote and laser pointer	50.00	10	500.00
2035	1	Everlast Hard Drive, 60 GB	200.00	6	1200.00
2036	1	Optical Mouse	45.00	30	1350.00
2037	1	Optical Mouse	45.00	10	450.00
2037	2	Roadster 56KB Ext. Modem	120.00	5	600.00
2037	3	Everlast Hard Drive, 60 GB	205.00	10	2050.00
2038	1	NoTech Speaker Set	50.00	8	400.00

Multidimensional View of Sales

Customer Dimension	Time Dimension		Totals
	15-May-08	16-May-08	
Dartonik	\$1,400.00	\$1,350.00	\$2,750.00
Summer Lake	\$1,800.00	\$3,100.00	\$4,900.00
Trydon		\$400.00	\$400.00
Totals	\$3,200.00	\$4,850.00	\$8,050.00

Sales are located in the intersection of a customer row and time column

Aggregations are provided for both dimensions

Advanced Database Support

- Advanced data access features include:
 - Access to many different kinds of DBMSs, flat files, and internal and external data sources
 - Access to aggregated data warehouse data
 - Advanced data navigation
 - Rapid and consistent query response times
 - Maps end-user requests to appropriate data source and to proper data access language
 - Support for very large databases

Easy-to-Use End-User Interface

- Advanced OLAP features more useful when access is simple
- Many interface features are “borrowed” from previous generations of data analysis tools
 - Already familiar to end users
 - Makes OLAP easily accepted and readily used

Client/Server Architecture

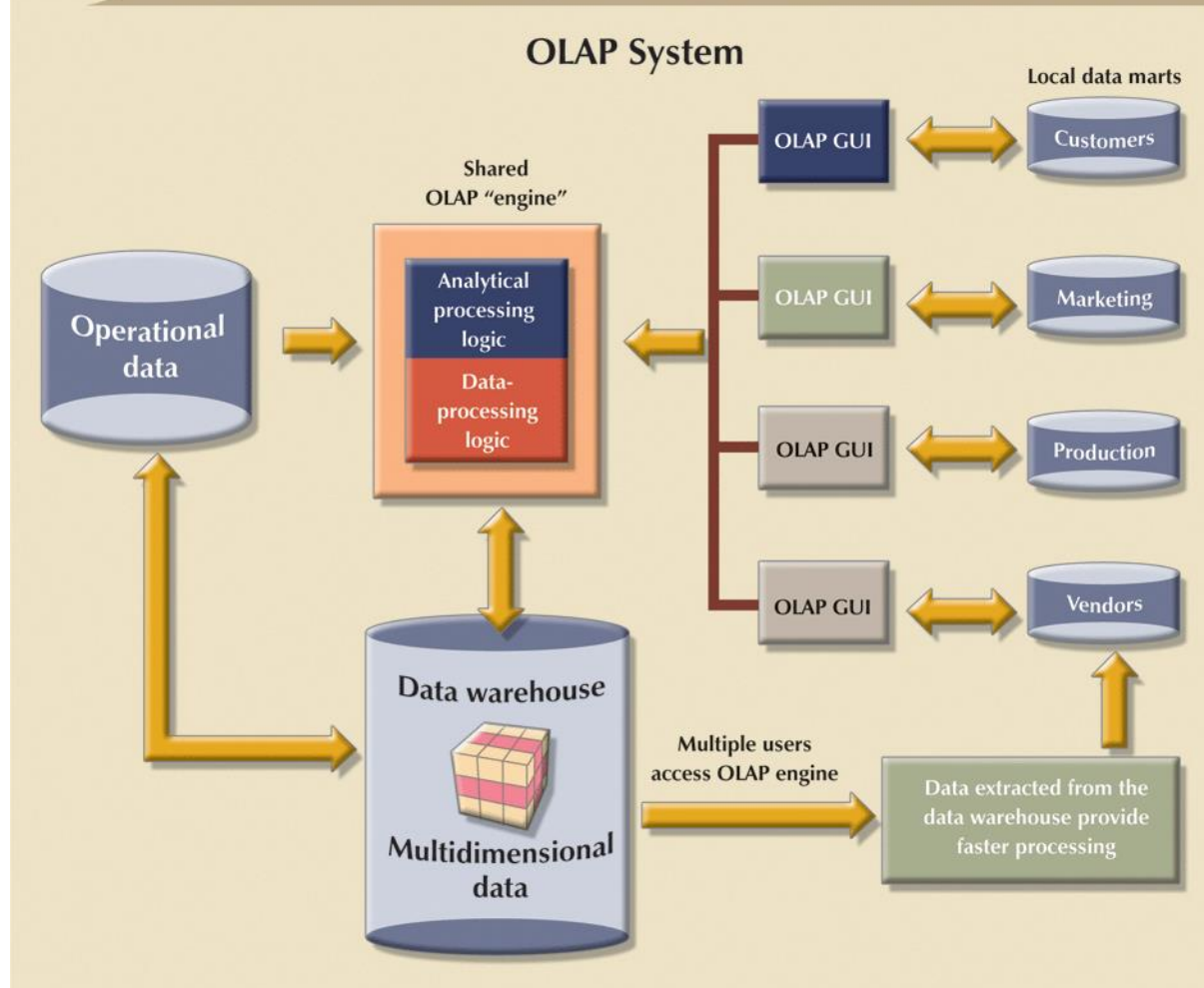
- Provides framework for design, development, implementation of new systems
 - Enables OLAP system to be divided into several components that define its architecture
 - OLAP is designed to meet ease-of-use as well as system flexibility requirements

OLAP Architecture

- Operational characteristics' three main modules:
 - Graphical user interface (GUI)
 - Analytical processing logic
 - Data-processing logic
- Designed to use both operational and data warehouse data
- In most implementations, data warehouse and OLAP are interrelated and complementary
- OLAP systems merge data warehouse and data mart approaches

FIGURE
13.10

OLAP server with local mini data marts



Relational OLAP

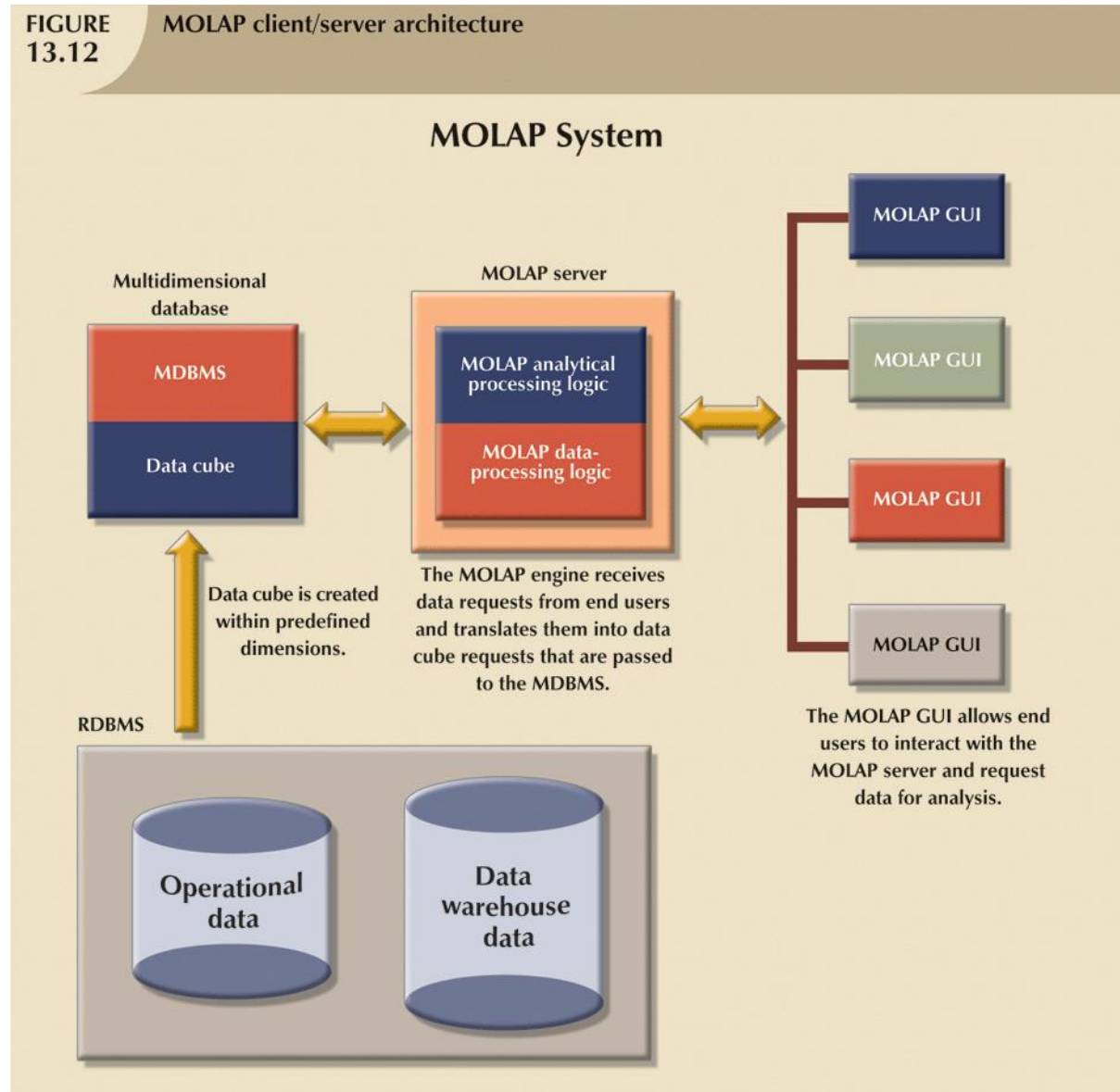
- Uses relational databases and relational query tools
 - Stores and analyzes multidimensional data
- Adds following extensions to traditional RDBMS:
 - Multidimensional data schema support within RDBMS
 - Data access language and query performance optimized for multidimensional data
 - Support for very large databases

Multidimensional OLAP

- Extends OLAP functionality to **multidimensional database management systems (MDBMSs)**
 - MDBMS end users visualize stored data as a 3D **data cube**
 - Data cubes can grow to n dimensions, becoming hypercubes
 - To speed access, data cubes are held in memory in a cube cache

FIGURE
13.12

MOLAP client/server architecture



Relational vs. Multidimensional OLAP

- Selection of one or the other depends on evaluator's vantage point
- Proper evaluation must include supported hardware, compatibility with DBMS, etc.
- ROLAP and MOLAP vendors working toward integration within unified framework
- Relational databases use star schema design to handle multidimensional data

Star Schema

- Data modeling technique
 - Maps multidimensional decision support data into relational database
- Creates near equivalent of multidimensional database schema from relational data
- Easily implemented model for multidimensional data analysis
 - Preserves relational structures on which operational database is built
- Four components: facts, dimensions, attributes, and attribute hierarchies

Facts

- Numeric measurements that represent specific business aspect or activity
 - Normally stored in fact table that is center of star schema
- **Fact table** contains facts linked through their dimensions
- **Metrics** are facts computed at run time

Dimensions

- Qualifying characteristics provide additional perspectives to a given fact
- Decision support data almost always viewed in relation to other data
- Study facts via dimensions
- Dimensions stored in **dimension tables**

Attributes

- Use to search, filter, and classify facts
- Dimensions provide descriptions of facts through their attributes
- No mathematical limit to the number of dimensions
- **Slice and dice**: focus on slices of the data cube for more detailed analysis

Attribute Hierarchies

- Provide top-down data organization
- Two purposes:
 - Aggregation
 - Drill-down/roll-up data analysis
- Determine how the data are extracted and represented
- Stored in the DBMS's data dictionary
- Used by OLAP tool to access warehouse properly

Star Schema Representation

- Facts and dimensions represented in physical tables in data warehouse database
- Many fact rows related to each dimension row
 - Primary key of fact table is a composite primary key
 - Fact table primary key formed by combining foreign keys pointing to dimension tables
- Dimension tables smaller than fact tables
- Each dimension record related to thousands of fact records

Performance-Improving Techniques for the Star Schema

- Four techniques to optimize data warehouse design:
 - Normalizing dimensional tables
 - Maintaining multiple fact tables to represent different aggregation levels
 - Denormalizing fact tables
 - Partitioning and replicating tables

Performance-Improving Techniques for the Star Schema (continued)

- Dimension tables normalized to:
 - Achieve semantic simplicity
 - Facilitate end-user navigation through the dimensions
- Denormalizing fact tables improves data access performance and saves data storage space
- Partitioning splits table into subsets of rows or columns
- Replication makes copy of table and places it in different location

Implementing a Data Warehouse

- Numerous constraints, including:
 - Available funding
 - Management's view of role played by an IS department
 - Extent and depth of information requirements
 - Corporate culture
- No single formula can describe perfect data warehouse development

The Data Warehouse as an Active Decision Support Framework

- Data warehouse:
 - Is not a static database
 - Is a dynamic framework for decision support that is always a work in progress
- Data warehouse is critical component of modern BI environment
- Design and implementation must be examined as part of entire infrastructure

A Company-Wide Effort That Requires User Involvement

- Data warehouse data cross departmental lines and geographical boundaries
- Building a data warehouse requires the designer to:
 - Involve end users in process
 - Secure end users' commitment from beginning
 - Create continuous end-user feedback
 - Manage end-user expectations
 - Establish procedures for conflict resolution

Satisfy the Trilogy: Data, Analysis, and Users

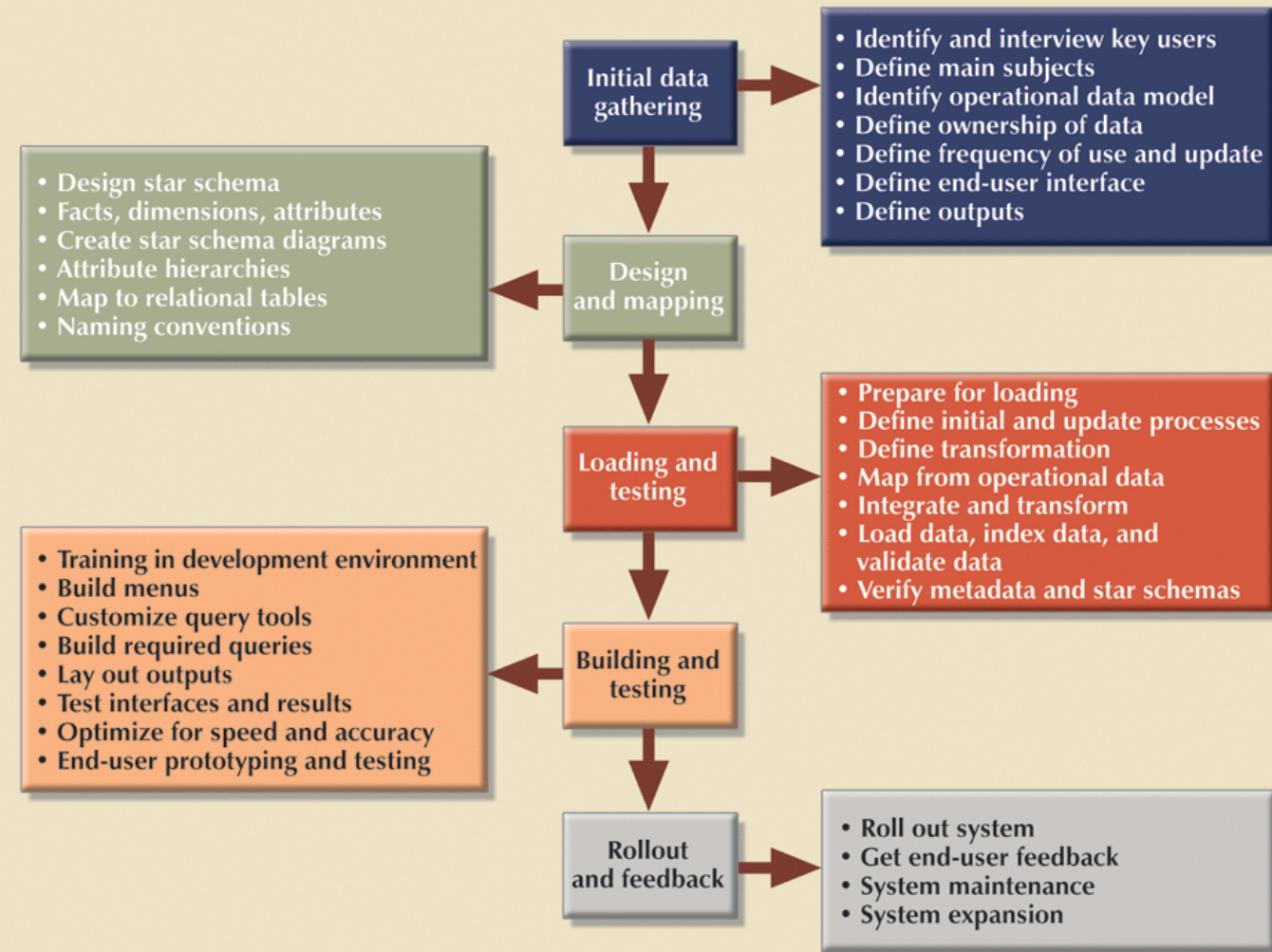
- Data warehouse designer must satisfy:
 - Data integration and loading criteria
 - Data analysis capabilities with acceptable query performance
 - End-user data analysis needs

Apply Database Design Procedures

- Company-wide effort requiring many resources
- Quantity of data requires latest hardware and software
- Detailed procedures to orchestrate flow of data from operational databases to data warehouse
- People with advanced database design, software integration, and management skills

FIGURE
13.22

Data warehouse design and implementation road map



Data Mining

- Data-mining tools do the following:
 - Analyze data
 - Uncover problems or opportunities hidden in data relationships
 - Form computer models based on their findings
 - Use models to predict business behavior
- Requires minimal end-user intervention

SQL Extensions for OLAP

- Proliferation of OLAP tools fostered development of SQL extensions
- Many innovations have become part of standard SQL
- All SQL commands will work in data warehouse as expected
- Most queries include many data groupings and aggregations over multiple columns

The ROLLUP Extension

- Used with GROUP BY clause to generate aggregates by different dimensions
- GROUP BY generates only one aggregate for each new value combination of attributes
- ROLLUP extension enables subtotal for each column listed except for the last one
 - Last column gets grand total
- Order of column list important

The CUBE Extension

- CUBE extension used with GROUP BY clause to generate aggregates by listed columns
 - Includes the last column
- Enables subtotal for each column in addition to grand total for last column
- Useful when you want to compute all possible subtotals within groupings
- Cross-tabulations good application of CUBE extension

Materialized Views

- A dynamic table that contains SQL query command to generate rows
 - Also contains the actual rows
- Created the first time query is run and summary rows are stored in table
- Automatically updated when base tables are updated

Summary

- Business intelligence generates information used to support decision making
- BI covers a range of technologies, applications, and functionalities
- Decision support systems were the precursor of current generation BI systems
- Operational data not suited for decision support

Summary (continued)

- Four categories of requirements for decision support DBMS:
 - Database schema
 - Data extraction and loading
 - End-user analytical interface
 - Database size requirements
- Data warehouse provides support for decision making
 - Usually read-only
 - Optimized for data analysis, query processing

Summary (continued)

- OLAP systems have four main characteristics:
 - Use of multidimensional data analysis
 - Advanced database support
 - Easy-to-use end-user interfaces
 - Client/server architecture
- ROLAP provides OLAP functionality with relational databases
- MOLAP provides OLAP functionality with MDBMSs

Summary (continued)

- Star schema is a data-modeling technique
 - Maps multidimensional decision support data into a relational database
- Star schema has four components:
 - Facts
 - Dimensions
 - Attributes
 - Attribute hierarchies

Summary (continued)

- Four techniques optimize data warehouse design:
 - Normalize dimensional tables
 - Maintain multiple fact tables
 - Denormalize fact tables
 - Partition and replicate tables
- Data mining automates analysis of operational data
- SQL extensions support OLAP-type processing and data generation